Q1 - 24 January - Shift 2

If $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$, $x\in R$, then

(1) $3f(1)+f(2)=f(3)$

(2) $f(3)-f(2)=f(1)$

(3) $2f(0)-f(1)+f(3)=f(2)$

(4) $f(1)+f(2)+f(3)=f(0)$

Space for your notes:

Q2 - 25 January - Shift 1

Let

$$y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$$

Then $y'-y''$ at $x=-1$ is equal to

(1) 976 (2) 464

(3) 496 (4) 944

Space for your notes:

Q3 - 29 January - Shift 1

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function that satisfies the relation $f(x+y)=f(x)+f(y)-1$, $\forall x, y\in\mathbb{R}$. If $f'(0)=2$, then $|f(-2)|$ is equal to _____.

Space for your notes:

Q4 - 29 January - Shift 2

Let f and g be twice differentiable functions on R such that

$f''(x) = g''(x) + 6x$

$f'(1) = 4g'(1) - 3 = 9$

$f(2) = 3g(2) = 12$

Then which of the following is NOT true ?

(1) $g(-2) - f(-2) = 20$

(2) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

(3) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(4) There exists $x_0 \in \left(1, \frac{3}{2}\right)$ such that $f(x_0) = g(x_0)$

Space for your notes:

Q5 - 31 January - Shift 1

Let

$$y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$$

. Then, at x = 1,

(1) $2y' + \sqrt{3}\pi^2 y = 0$

(2) $2y' + 3\pi^2 y = 0$

(3) $\sqrt{2}y' - 3\pi^2 y = 0$

(4) $y' + 3\pi^2 y = 0$

Space for your notes:

Q6 - 01 February - Shift 2

If $y(x) = x^x$, $x > 0$, then $y''(2) - 2y'(2)$ is equal to

(1) $8 \log_e 2 - 2$

(2) $4 \log_e 2 + 2$

(3) $4 (\log_e 2)^2 - 2$

(4) $4 (\log_e 2)^2 + 2$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3) **Q2** (3) **Q3** (3) **Q4** (2)

Q5 (2) **Q6** (3)

Q1 (3)

$$f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3),\ x\in R$$

Let $f'(1)=a,\ f''(2)=b,\ f'''(3)=c$

$$\mathbf{f(x)=x^3-ax^2+bx-c}$$

$$f'(x)=3x^2-2ax+b$$

$$f''(x)=6x-2a$$

$$f'''(x)=6$$

$c=6,\ a=3,\ b=6$

$f(x)=x^3-3x^2+6x-6$

$f(1)=-2,\ f(2)=2,\ f(3)=12,\ f(0)=-6$

$2f(0)-f(1)+f(3)=2=f(2)$

Q2 (3)

$$y=\frac{1-x^{32}}{1-x}\Rightarrow y-xy=1-x^{32}$$

$$y'-xy'-y=-32x^{31}$$

$$y''-xy''-y'-y'=-(32)(31)x^{30}$$

at $x=-1\Rightarrow y'-y''=496$

Q3 (3)

$f(x+y) = f(x) + f(y) - 1$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = f'(0) = 2$$

$f'(x) = 2 \Rightarrow dy = 2dx$

$y = 2x + C$

$x = 0, y = 1, c = 1$

$y = 2x + 1$

$$|f(-2)| = |-4+1| = |-3| = 3$$

Q4 (2)

$f''(x) = g''(x) + 6x \quad \ldots(1)$

$f'(1) = 4g'(1) - 3 = 9 \quad \ldots(2)$

$f(2) = 3g(2) = 12 \quad \ldots(3)$

By integrating (1)

$$f'(x) = g'(x) + 6\frac{x^2}{2} + C$$

At $x = 1$,

$f'(1) = g'(1) + 3 + C$

$\Rightarrow 9 = 4 + 3 + C \Rightarrow C = 3$

$\therefore \; f'(x) = g'(x) + 3x^2 + 3$

Again by integrating,

$$f(x) = g(x) + \frac{3x^3}{3} + 3x + D$$

At $x = 2$,

$f(2) = g(2) + 8 + 3(2) + D$

$\Rightarrow 12 = 4 + 8 + 6 + D \Rightarrow D = -6$

So, $f(x) = g(x) + x^3 + 3x - 6$

$\Rightarrow f(x) - g(x) = x^3 + 3x - 6$

At $x = -2$,

$\Rightarrow g(-2) - f(-2) = 20$ (Option (1) is true)

Now, for $-1 < x , 2$

$h(x) = f(x) - g(x) = x^3 + 3x - 6$

$\Rightarrow h'(x) = 3x^2 + 3$

$\Rightarrow h(x)\uparrow$

So, $h(-1) < h(x) < h(2)$

$\Rightarrow -10 < h(x) < 8$

$\Rightarrow |h(x)| < 10$ (option (2) is NOT true)

Now, $h'(x) = f'(x) - g'(x) = 3x^2 + 3$

If $|h'(x)| < 6 \Rightarrow |3x^2 + 3| < 6$

$\Rightarrow 3x^2 + 3 < 6$

$\Rightarrow x^2 < 1$

$\Rightarrow -1 < x < 1$ (option (3) is True)

Q5 (2)

$y = \sin^3(\pi/3 \cos g(x))$

$$g(x) = \frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{3/2}$$

$g(1) = 2\pi/3$

$$y' = 3\sin^2\left(\frac{\pi}{3}\cos g(x)\right) \times \cos\left(\frac{\pi}{3}\cos g(x)\right) \times \frac{\pi}{3}\left(-\sin g(x)\right) g'(x)$$

$$y'(1) = 3\sin^2\left(-\frac{\pi}{6}\right).\cos\left(\frac{\pi}{6}\right).\frac{\pi}{3}\left(-\sin\frac{2\pi}{3}\right) g'(1)$$

$$g'(x) = \frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{1/2}\left(-12x^2 + 10x\right)$$

$$g'(1) = \frac{\pi}{2\sqrt{2}}\left(\sqrt{2}\right)(-2) = -\pi$$

$$y'(1) = \frac{\cancel{3}}{4}.\frac{\sqrt{3}}{2}.\frac{\pi}{\cancel{3}}\left(\frac{-\sqrt{3}}{2}\right)(-\pi) = \frac{3\pi^2}{16}$$

$$y(1) = \sin^3\left(\pi/3\cos 2\pi/3\right) = -\frac{1}{8}$$

$$2y'(1) + 3\pi^2 y(1) = 0$$

Q6 (3)

$y' = x^x$

$y' = x^x (1 + \ell nx)$

$y'' = x^x (1 + \ell nx)^2 + x^x \cdot \frac{1}{x}$

$y'' (2) = 4(1 + \ell n2)^2 + 2$

$y' (2) = 4(1 + \ell n2)$

$y'' (2) - 2y'(2) = 4(1 + \ell n2)^2 + 2 - 8(1 + \ell n2)$

$= 4(1 + \ell n2) [1 + \ell n2 - 2] + 2$

$= 4(\ell n2)^2 - 1) + 2$

$= 4(\ell n2)^2 - 2$